Introduction

Bounded Research As Infrastructure I

AutoResearch systems are most useful when their planning, evidence, evaluation, and review surfaces remain inspectable. The recent pattern popularized by bounded coding-agent research loops is simple: define a tractable objective, try candidate changes under a budget, keep the result that improves the metric, and leave a replayable trace of what happened [@karpathy_autoresearch_2026]. That pattern is powerful, but it is also easy to overstate. A public research template should show how to run the loop without hiding cost, evidence, review, or execution boundaries.

Exemplar Task I

This project implements that safer version. The central task is `small MNIST neural-network classification`: `MNIST handwritten digit database` from the handwritten-digit database [@lecun_mnist_database], a `nearest-centroid` baseline, and a finite list of candidate model families (`MLP, nearest-centroid, softmax regression, tiny patch-attention`). The AutoResearch loop is responsible for proposing candidate configurations, evaluating them against `test_accuracy`, selecting the best result with deterministic tie-breaking, and writing the evidence needed to review the claim.

Contribution And Boundary I

The contribution is not a new `MNIST` classifier. It is a template-level demonstration of how bounded AutoResearch can be orchestrated through the same lifecycle used for reproducible papers: tests run first, analysis writes structured artifacts, rendering hydrates manuscript variables, validation checks evidence and readiness, and copy stages publish final deliverables. The default path makes no network calls, no LLM calls, executes no generated code, and never treats a generated review packet as human approval. The methods in @sec:methodology define that boundary, and the results in @sec:results report only what the validated artifacts support.

Process Organization Motif I

The exemplar also treats research orchestration itself as a first-class research object. In that limited operational sense, “research for research’s sake” means that programs, ledgers, evidence registries, validation gates, and manuscript hydration do more than report a result: they maintain the conditions under which the next research step can be inspected, replayed, criticized, and extended. This is a process analogy, not a moral claim that software is an end in itself or a biological claim that the template is alive.

Related Work And Current Trends I

Related Work And Current Trends II

Information Overload And Machine-Readable Science

The current AutoResearch literature is driven by a practical bottleneck: scientific output is growing faster than document-centered review and synthesis can absorb. Recent science-of-science evidence complicates any simple productivity story: AI-augmented scientists can publish and be cited more often, but the same adoption pattern can narrow the collective range of topics and interactions in science [@hao_ai_tools_focus_2026]. This is a 2026 Nature result, not a 2024 analysis, and it motivates governance rather than celebration.

One response is to make literature synthesis more source governed. OpenScholar uses retrieval-augmented generation over a large scientific passage store and reports citation accuracy improvements on literature synthesis tasks [@asai_openscholar_2026]. PaperQA2 similarly evaluates literature-search agents against expert scientific tasks and emphasizes cited answers, contradiction detection, and factuality [@skarlinski_paperqa2_2024]. STORM, PaperQA, and GPT Researcher are adjacent source-grounded writing systems that motivate this project's insistence on citation-backed claims and